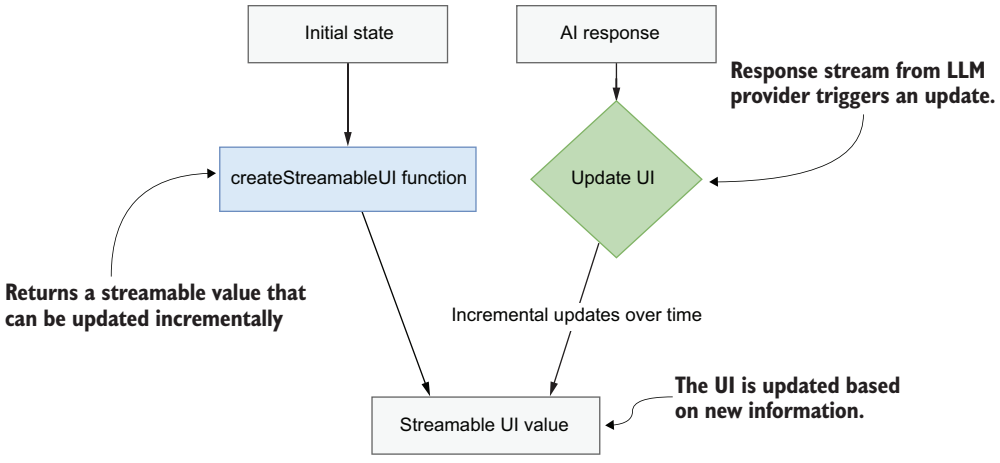


Figure 4.7 `createStreamableUI` enables the creation of UI components that can be updated incrementally. This capability enables real-time modifications to the user interface as new information is received from the LLM provider.

- `append`—Adds new content to the existing streamable UI
- `error`—Handles any errors that occur during the streaming process or displays an error component

The `createStreamableUI` function in the Vercel AI SDK provides users with immediate updates as processes complete, allowing incremental updates to the UI.

In this example, we create a streamable UI, update it with initial content, append new content as steps are completed, and finally mark it as done. If an error occurs, we use the `error` method to handle it. This demonstrates how `createStreamableUI` can be used to walk a user through a two-step process, informing them when each step is complete.

Listing 4.5 Adding incremental UI updates to the stream

This initializes a new streamable UI that we can update incrementally.

We use the update method to set the initial content of our UI.

The append method adds new content without replacing the existing content.

```
import { createStreamableUI } from '@vercel/ai-sdk';
export async function continueConversation(history) {
  const ui = createStreamableUI();
  ui.update(<p>Starting process...</p>);
  try {
    await new Promise(resolve => setTimeout(resolve, 1000));
    ui.append(<p>Step 1 complete</p>);

    await new Promise(resolve => setTimeout(resolve, 1000));
    ui.append(<p>Step 2 complete</p>);
```